



Daffodil International University
Faculty of Science & Information Technology
Department of Software Engineering
Mid Term Examination, Spring-2025
Course Title: Bangladesh Studies, Course Code: BNS 101
Batch: 44, Teachers: DEH, MSI, TS

Exam Duration: 1.30 Hours

Marks: 25

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1	Answer the following questions briefly: a. Examine opportunities and threats of Bangladesh's geographical location. b. Illustrate Bangladesh's key cultural festivals. c. Examine globalization's impact on Bangladeshi culture. d. Demonstrate key challenges for Bangladesh's demographic dividend. e. Classify organs or branches of the 'Government'	(5x2) =10	CLO-1 L-3
2	Analyze the challenges Bangladesh faces in maintaining relationship with India, in line with the fundamental principle of its foreign policy— "friendship to all, malice towards none."	5	CLO-2 L-4
3	Explain how the concept of 'mixed blood' shaped the Bengali national identity.	5	CLO-1 L-2
4	Evaluate the July 2024 Mass Uprising considering the fundamental rights— 'freedom of speech' and 'freedom of assembly'— in the Bangladesh constitution.	5	CLO-2 L-5



Daffodil International University

Department of Software Engineering

Faculty of Science & Information Technology

Midterm Examination, Spring 2025

Course Code: SE 111; Course Title: Computer Fundamentals

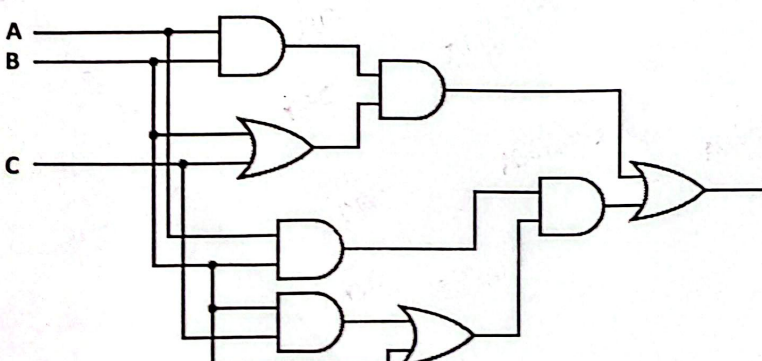
Sections & Teachers: NJM(A,B), MRD(C,D), SI(E,I,J), MKS(F), TAK(G,H), KR(K,M), KF(L)

Time: 1 Hour 30 Mins

Marks: 25

Answer ALL Questions

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1.	a) TechNova a small company recently installed operating system for their office computers. One of the characteristics of that operating system is, it is UNIX operating system with C programming. UNIX is a multi-user and multitasking operating system designed for efficiency and flexibility. The main reason to install this type of operating system is, it supports multiple users at once and provides a command-line interface Explain the generation of computer mentioned in the above scenario with its other software technologies.	[Marks-3]	CLO-1 C2
	b) A computer runs a program instantly accessing data from a type of storage for quick processing. Later, it saves files sequentially for long-term use in another type of storage. When needed again the system searches in order to retrieve the stored data. This ensures fast execution for active tasks while keeping information accessible for future use. Compare the above-mentioned memories with example.	[Marks-3]	
	c) i) Convert 456_9 to its Base 4 equivalent Number. ii) Convert 1111110.1001_2 to its Base 10 equivalent Number. iii) Convert $D16_{16}$ to its Base 8 equivalent Number.	[Marks-9]	
2.	a) Illustrate a logic circuit for the following Boolean expression. $F = AB'(AC+BC)' + A'C + A'B'(BC') + AC + B'C$	[Marks-3]	CLO-2 C4
	b) Identify the final output for the below logic circuit diagram also simplify the output result and prove your simplification using truth table. 	[Marks-7]	



Daffodil International University

Department of Software Engineering

Faculty of Science & Information Technology

Midterm Examination, Spring 2025

Course Code: ENG-101; Course Title: English-I

Sections & Teachers: All Sections; MA, JC, SIS

Time: 1 Hour 30 Mins

Marks: 25

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

Read the Passage Carefully and Answer the Questions that Follow:

Sleeping On The Job

North Americans are a group who do not take naps. Generally, Afternoon naps denote inefficiency and laziness. Latin American and some European cultures have a similar perspective. Taking naps during noon and 4:00 p.m. is common in nearby companies of Mexico and Greece. Recent research suggests that 15 - 30 minutes rest at work in the evening will serve employees to be more conscious, energetic and happier in what they are doing. They also become optimistic and carry the task more confidently. Still, napping on the work has not become normalised. In the academic world, there is a serious debate happening on the topic of “power napping” benefits.

Some data reports that by estimating the standard of eight hours of sleep per night, the average American has 500 hours of sleep debt or sleep deficit. The National Sleep Foundation in America conducted a survey in which it is found that two out of every three Americans have less than 7 hours of sleep per night during the business week. 40 percent of people claim that fatigue disturbs their everyday lives. Exhaustion is considered as a major issue in contemporary culture by William Anthony, a professor of psychiatry at Boston University. He claims that after drunkenness, sleeping is the leading cause of car crashes. Professor Anthony says that “We have a clear message”. Citizens should be allowed to sleep after their breaks. It is reasoned based on productivity. They will not be efficient if they are tired and asleep on the job.

Majorly for safety reasons, most employers motivate employees to sleep at work. The Metropolitan Transportation Authority, which runs both the New York subway system and the two commuter railroads, is permitting its bus and train technicians to take power sleep. One more railway starts to allow its train drivers to sleep about 45 minutes, during which trains are delayed when specified off-main-line locations and when emergency responders have been alerted. Some international airlines permit airline pilots to sleep in the cockpit, when they are not on service. United States’ airline companies still have not agreed to this procedure.

According to the Encyclopedia of Sleep and Dreaming, taking naps in mid afternoon is regarded as a biologically-based inclination and so the habit of sleeping at night is formed. If the required sleep limit is disrupted or disturbed for some reason, a nap acts as an adjustment. Additionally, it could reduce sleepiness and boost metabolism. Many experiments suggest that the nap zone occurs between midday and 3:00 p.m. To solve this problem, several people use caffeine or sugar. But, the employee's mood and efficiency will improve, if they are permitted to take naps.

Workers could have the ability to focus better and complete assignments without taking a single day out. Still, Workers take naps without authorization, though some businesses started to implement the naps for employees to increase productivity. One of the US companies is constructing a 2,000 square foot building for nap which can be filled upto 20 of its employees at any given time. Even a Japanese firm in their corporate offices, sets up tents and gives eyeshades and earplugs and permits workers to take naps during the workday. According to Professor Anthony, napping will not be one of the operational activities. It will have a great influence on productivity. Smart managers identified that workers need rest in order to perform well in their work. There is also a belief that company naptime is just a way to keep employees to work for longer periods of time. On the other hand, nap days come so frequently as work hours become more flexible for certain employees. Office hours are being there till employees awake. About 11 million Americans are teleworking and forty million spending bonus full or part time. Production hours are also increasing which takes their toll. As a consequence, Americans sleep little and work more hours. By understanding this, companies arrange quick naps for employees in the middle of the day for the benefit of the companies.

1.	a)	Classify the following statements as Yes/No/Not Given based on your reading of the passage. i. Recent research says that power napping does not have any impact on employees. ii. Americans have a sleep debt of 500 hours. iii. Power Naps will increase the age. iv. One US company built a 2,000-square-foot building for <u>play</u> .	[4x1=4]	CLO- 3 Level- C4
	b)	Select the correct information from the passage to complete the sentences below. i. Citizens should be allowed to sleep after their ____ which will increase their productivity. ii. Employees' mood and ____ will improve if they are allowed to take nap. iii. A nap can reduce sleepiness and boost ____ iv. Smart managers recognize that workers need ____ to perform better in their work. v. Nap days come so frequently as work hours become more ____ for certain employees. vi. American employees take a nap at midday to ____ their companies.	[6x0.5=3]	

2.	a) Identify an appropriate article (a/an/the/put cross for no article) to fill in the gaps. During (i) _____ summer vacation, I visited (ii) _____ beautiful country called (iii) _____ Switzerland. I stayed in (iv) _____ charming village nestled in (v) _____ Alps. Every morning, I enjoyed a fresh coffee and a croissant from (vi) _____ local bakery. One day, I decided to explore (vii) _____ nearby lake that was known for its crystal-clear water. I rented (viii) _____ small boat and spent the afternoon paddling around, admiring (ix) _____ breathtaking scenery. By the end of my trip, I felt refreshed and inspired by (x) _____ serene environment.	[10x0.5=5] CLO- 1 Level- C1
	b) Write the correct form of the verb in the brackets to complete the following sentences. i. We _____ (wait) for the bus for over an hour before it finally arrived. ii. She _____ (not visit) her grandparents last weekend. iii. I usually _____ (go) to the gym after work. iv. At 7 p.m. yesterday, they _____ (watch) TV. v. We _____ (live) in this city for ten years.	[5x1=5]
3.	Construct a paragraph on the following topic Your Neighborhood Or A frightening/ dangerous experience.	[1x8=8] CLO- 4 Level- C3



Daffodil International University
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Midterm Examination, Spring 2025; Course Code: SE 113

Course Title: Introduction to Software Engineering

Sections & Teachers: A, B, C, D, E, F, G, H, I, J, K, L, M & SS, MKS, NS, KR, LA, SA

Time: 1:30 Hrs

Marks: 25

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1.	a)	Explain which characteristics are common in a high quality software with reasons.	[Marks-4]	CLO-1 Level-2
	b)	Briefly discuss the basic responsibilities of a software engineer that are essential to develop software.	[Marks-4]	
	c)	Predict an appropriate career that you may like to have in near future with reasons.	[Marks-3]	
	d)	Distinguish between plan-driven and agile processes	[Marks-3]	
	e)	Differentiate between the waterfall model and incremental model with examples.	[Marks-4]	
2.	a)	The upcoming software project to serve the Non Residential Bangladeshi citizens would be very time critical as well as hard to be satisfactory because of high variety of scopes – some of which are still not clear. As the foreign ministry is really eager to complete the project with high satisfaction of all the stakes which is a bit risky, hence the ministry likes to provide flexibility to the development team with budget and working environment. Determine a related Software Development Life Cycle model for the above-mentioned issues.	[Marks-3]	CLO-2 Level-3
	b)	Derive a SDLC model with each phase that relate the suitability for case given in scenario 2.a)	[Marks-4]	